Assignment 1: Schema Matching[1]

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Abstract. *The goal of this assignment is to find the correspondences between columns in two datasets, which is called schema matching.*

Submission Guidelines

Submission Deadline: 03.11.2025 **Presentation:** 05.11.2025

How to submit:

- Submit a compressed file that contains (1) the PDF report and (2) your source code files via **ISIS**.
- One submission per each group is enough.
- You are free to use whatever tools/programming languages you intend to.
- Ensure that both group members' names are clearly listed at the beginning of your final PDF report.

1. Setup

1.1. Datasets

We have two relational datasets with the following schemas:

$$imdb(Id, Name, YearRange, ReleaseDate, Director, Creator, Cast, Duration, RatingValue, ContentRating, Genre, Url, Description),$$
$$rotten_tomatoes(Id, Name, Year, Release\ Date, Director, Creator, Actors, Cast, Language, Country, Duration, RatingValue, RatingCount, ReviewCount, Genre, Filming\ Locations, Description).$$

The datasets have been attached to this document.

1.2. Ground Truth

The set of actual correspondences is as follows:

$$G = \{\langle Imdb.Name, rt.Name\rangle, \langle Imdb.YearRange, rt.Year\rangle, \langle Imdb.ReleaseDate, rt.\text{"}Release\ Date\text{"}\rangle, \langle Imdb.Director, rt.Director\rangle, \langle Imdb.Creator, rt.Creator\rangle, \langle Imdb.Cast, rt.Cast\rangle, \langle Imdb.Duration, rt.Duration\rangle, \langle Imdb.RatingValue, rt.RatingValue\rangle, \langle Imdb.Genre, rt.Genre\rangle, \langle Imdb.Description, rt.Description\rangle\}.$$

Overall, we have 10 pairs of columns that are corresponded.

1.3. Evaluation Measures

In the following tasks, you should evaluate the correctness and completeness of your implemented solutions in terms of precision and recall:

$$precision = \frac{Number\ of\ the\ discovered\ correspondences\ that\ are\ in\ G}{Number\ of\ all\ the\ discovered\ correspondences},$$
$$recall = \frac{Number\ of\ the\ discovered\ correspondences\ that\ are\ in\ G}{Number\ of\ all\ the\ actual\ correspondences = 10}.$$

2. Tasks

2.1. Label-Based Schema Matching

Here, we want to find the correspondences between the columns from the two datasets with the help of only *schema headers*.

1. Provide an algorithm. Specify the input, output, similarity function, and time complexity.
2. Implement the algorithm and report the results. Is there any parameter that affects the results?
3. What is the upsides and downsides of this method? When does it work and when not?

2.2. Instance-Based Schema Matching

Here, we want to find the correspondences between the columns from the two datasets with the help of only *data values*.

1. Provide an algorithm. Specify the input, output, similarity function, and time complexity.
2. Implement the algorithm and report the results. Is there any parameter that affects the results?
3. What is the upsides and downsides of this method? When does it work and when not?

2.3. Instance-based Schema Matching using Large Language Models

In this task, we aim to identify correspondences between columns from two datasets using only *data values* with the assistance of a Large Language Model (LLM).
Hint: You are free to choose any large language model available to you. One option is to use platforms such as Ollama [1], which provide access to various models.

1. Design your solution so that it relies exclusively on the data values from the datasets for matching.
2. Provide detailed descriptions of your prompt(s) used with the LLM, explaining how they are constructed to solve the task.
3. Report precision and recall of the schema matching results, inference time for the LLM to generate predictions, and the amount of data used to achieve the reported results.
4. Compare your results with those presented in Section 2.2. Discuss the strengths and limitations of using this instance-based approach with LLMs in contrast to the method you used in Section 2.2.

[1] `https://ollama.com`